

D is incorrect

At Y, equilibrium position, the air molecule has the highest speed.

18 C

$$f = \frac{v}{d}$$

$$10 = \frac{2.0}{d}, \quad d = 0.20 \text{ m}$$

$$\lambda = 2d = 0.40 \text{ m}$$

Distance moved from initial position at M to second minimum intensity = $3(1/4 \lambda) = 0.30 \text{ m}$

19 A

$$1 - \frac{1}{2} - \frac{1}{2}$$